# VANTARA RETAIL SOLUTIONS

## Data Science & Analytics Division

### Final Technical Report — Customer Behavior Prediction Platform

> Rebuilt against the approach statement and PRD acceptance criteria, using **out-of-time** evaluation. Headline numbers are lower than the original draft because the original random split leaked future information; §10 explains the difference. All tables are regenerated from MLflow / the model card by `scripts/make_report_tables.py`.

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## 1. Executive Summary & Business Context

Vantara Retail Solutions operates an omnichannel retailer serving consumers and small resellers. The Retention and Marketing teams previously relied on reactive, rule-based segments ("no order in 90 days") that treat every customer in a segment identically. This platform replaces that with a customer-level decision system:

1. **Calibrated churn risk** — out-of-time test **ROC-AUC 0.756**, recall **75.8%** at the operating threshold (recall ≥ 0.70).
2. **90-day customer value** — **LightGBM Tweedie**, validation **MAE £284**; it captures **63.9%** of the future value in the top decile (baseline: 12.5%).
3. **Value-at-risk prioritisation** — the at-risk leaderboard is ranked by `calibrated P(churn) × predicted 90-day value`, available in the API and dashboard.
4. **Explainable retention recommendations** — a rule table maps (segment, risk tier, value tier, top SHAP drivers) to an action, each with a cited reason.
5. **Probabilistic CLV benchmark (Beyond PRD)** — BG/NBD + Gamma-Gamma, with P(alive), reported honestly against the supervised model.
6. **Budget-constrained targeting (Beyond PRD)** — a greedy knapsack planner that maximises expected retained value per £ spent, with a save-rate sensitivity grid.

**Stack:** Python 3.11, pandas, scikit-learn, XGBoost, LightGBM, PyTorch, SHAP, LIME, Optuna, MLflow, pandera, lifetimes, FastAPI, SQLAlchemy 2, PostgreSQL 15, Streamlit, Docker Compose, pytest, ruff, GitHub Actions.

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## 2. Data Engineering & Leak-Free Architecture

- **Source:** UCI *Online Retail II* (`id=502`), 1,067,371 line items, 2009-12-01 → 2011-12-09.
- **Cleaning:** drop missing Customer ID for customer-level modelling, deduplicate exact lines, remove administrative stock codes; returns/cancellations are **flagged, not dropped** (return behaviour is a signal). Cleaned corpus cached to Parquet.
- **Point-in-time discipline:** for every cutoff `T`, features use only rows with `invoice_date < T`; labels use only the window `[T, T + 90d)`. This is asserted in tests.
- **Active cohort:** a customer enters the retention cohort only with a purchase in `[T − 365d, T)` (`active_lookback_days: 365`). Customers who left long ago are not a retention target and previously inflated the churn rate.

### 2.1 Stacked monthly snapshots (plan D1)

Ten monthly training cutoffs (2010-06-01 … 2011-03-01) are stacked, giving **36,454 training rows** rather than ~3.5k; validation is the 2011-06-01 snapshot and the test snapshot is 2011-09-01, **scored once**. Every training label window closes before the validation cutoff, and every validation window before the test cutoff.

| Snapshot | Cutoffs | Rows | Churn rate |
|---|---|---|---|
| Train | 2010-06-01 … 2011-03-01 (10) | 36,454 | 41%–67% |
| Validation | 2011-06-01 | 4,325 | 63.4% |
| Test (once) | 2011-09-01 | 4,323 | 50.5% |

*Cutoff 2011-09-01:* observation rows 623,150; label-window rows 151,811; cohort 4,323.

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## 3. Supervised Model Leaderboards

### 3.1 Churn classification (selection on the validation snapshot)

Hyperparameters were tuned with **Optuna (TPE)** scored by forward-chaining CV over the training cutoffs only. The model is chosen on validation ROC-AUC; the test snapshot is scored once.

| Model | Val ROC-AUC | Val Recall | Val Precision | Val F1 | CV ROC-AUC (fwd-chain) |
|---|---|---|---|---|---|
| **Logistic Regression (production)** | **0.7967** | 0.755 | 0.810 | 0.782 | 0.7815 |
| Random Forest | 0.7897 | 0.814 | 0.794 | 0.804 | 0.8347 |
| Decision Tree | 0.7873 | 0.804 | 0.800 | 0.802 | 0.7740 |
| XGBoost | 0.7617 | 0.808 | 0.770 | 0.788 | 0.8869 |
| LightGBM | 0.7602 | 0.781 | 0.781 | 0.781 | 0.8613 |
| KNN | 0.7562 | 0.816 | 0.770 | 0.792 | 0.7976 |

**Calibration and threshold (plan D4):** isotonic regression is fit on validation; the operating threshold is the *highest-precision* threshold that keeps recall ≥ 0.70 → **threshold 0.6709** (validation recall 0.708, precision 0.828). Risk tiers come from quantiles of the calibrated probability, not hand-picked constants.

**Out-of-time test result (used once):** ROC-AUC **0.7559**, recall **0.7579**, precision 0.6722, F1 0.7125, accuracy 0.6914, calibrated Brier 0.2143. AUC is below the PRD's 0.80 target; §10 explains why and why the number is nonetheless trustworthy.

### 3.2 90-day value regression (selection on validation MAE + value capture)

Every model must beat the **momentum baseline** (spend in the last 90 days). Selection is on validation MAE, tie-broken by top-decile value capture.

| Model | Val $R^2$ | Val MAE (£) | Val RMSE (£) | Top-decile value capture |
|---|---|---|---|---|
| **LightGBM Tweedie (production)** | 0.7543 | **283.76** | 1065.09 | **0.6391** |
| Random Forest | 0.7986 | 314.04 | 964.24 | 0.6386 |
| Spend Baseline (last 90d) | –0.0300 | 372.06 | 2180.61 | 0.1253 |
| Ridge (log1p) | –2.6231 | 442.34 | 4089.81 | 0.5840 |

The production model beats the baseline by ~24% MAE and captures **5×** more value in the top decile. Out-of-time test: $R^2$ 0.4887, MAE £437.94, top-decile capture 0.5798.

### 3.3 Probabilistic CLV benchmark (Beyond PRD P1)

BG/NBD (order frequency) + Gamma-Gamma (spend per order) on the same observation window, with **P(alive)** per customer:

| Model | Customers | MAE (£) | Top-decile capture |
|---|---|---|---|
| **Supervised LightGBM Tweedie** | 3,456 | ~284 | ~0.64 |
| BG/NBD + Gamma-Gamma | 1,656 | 524.92 | 0.403 |

**Honest negative:** the textbook probabilistic model does **not** beat the supervised model here, because the cohort is small and the 90-day horizon is short. Its value is the independent **P(alive)** view, shown in Customer 360.

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## 4. Deep Learning Comparative Analysis (out-of-time test snapshot)

| Architecture | Objective / Input | Test metric | Finding |
|---|---|---|---|
| Feed-Forward ANN | Tabular features, BatchNorm, Dropout(0.25), early stopping | ROC-AUC **0.7492**, recall **0.9124** | Highest recall; useful as an ensemble consensus check. |
| Sequence LSTM | Ordered `[gap_days, spend, qty]` per customer | ROC-AUC **0.7765**, recall **0.9052** | Learns temporal cadence beyond static features; matches the best classical model. |
| Spending Autoencoder | Unsupervised reconstruction, 17→12→4→12→17 | threshold **0.7487** (95th pct) | Flags ~5% of accounts for manual review. |

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## 5. Customer Segmentation & Persona Profiles

K-Means (k=4, silhouette **0.4618**, Davies-Bouldin **0.7585**) with GMM as the alternative-algorithm check (silhouette 0.1603). Personas are assigned by a rank-based rule so no two clusters collapse to the same label.

| Persona | n | Avg recency | Avg freq | Avg spend | Churn rate | Recommended play |
|---|---|---|---|---|---|---|
| High-Value Champions (VIP) | 20 | 24 d | 126.3 | £134,398 | 5% | Loyalty / early access, no discount |
| Loyal & Active Frequent Buyers | 2,328 | 56 d | 8.8 | £3,552 | 34% | Cross-sell, category expansion |
| High-Return Service-Recovery | 57 | 177 d | 3.7 | £1,739 | 72% | Service-recovery outreach, no discount |
| Dormant / Low-Engagement | 1,918 | 251 d | 2.5 | £755 | 71% | Automated win-back email |

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## 6. Retention Decision System

### 6.1 Value at risk (plan D4)

`value_at_risk = calibrated P(churn) × predicted 90-day value`. The batch scorer persisted **4,269 scored customers**, total value at risk ≈ **£412,835**.

### 6.2 Recommendation engine (plan D5)

`config/recommendations.yaml` maps (risk tier, value tier, top SHAP drivers, segment) to an action. First match wins; every action cites its trigger. There is **no treatment/outcome history**, so learning which intervention works (uplift modelling) is impossible — the rules encode policy, and §10 proposes the A/B test to replace the assumed save rate with a measured one.

### 6.3 Budget-constrained targeting (Beyond PRD P3)

Given budget $B$, channel costs $c$, and an assumed save rate $s$, a greedy knapsack selects the customers maximising $\Sigma\ p \cdot value \cdot s - c$ subject to $\Sigma\ c \le B$. Greedy is validated against a brute-force optimum in the tests. Because $s$ is unmeasured, the API returns a **sensitivity grid** over $s \in \{5\%, 10\%, 20\%\}$ and a four-policy comparison (top-risk / top-spender / rule-table / optimised). *Save-rate values are assumptions and are labelled as such.*

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## 7. Model Explainability (XAI)

- **Global:** SHAP summary in `docs/shap_global_summary.png`; recency and engagement dominate.
- **Local:** per-customer SHAP **and LIME** are served by the API (`GET /customers/{id}/explain`) on the served model's stored feature vector, and shown side by side in the dashboard (PRD acceptance criterion 4).
- **Plain language:** a narrative template translates the drivers into a sentence a marketing analyst can read, e.g. *"Calibrated churn risk 43.9% (Moderate). Top warning sign: recency."*

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## 8. Deployment, Monitoring & Reproducibility

- **One command:** `docker compose up --build` starts PostgreSQL, a one-off **scorer** (populates predictions/segments/recommendations), the **API** (:8000) and the **Streamlit dashboard** (:8501). Credentials come from `.env` (no hard-coded secrets).
- **Pipeline:** `python -m src.pipeline all` runs ingest → clean → validate → snapshots → train → segment → explain → score. `dvc.yaml` declares the same stages.
- **Serving:** the API loads artifacts in a FastAPI `lifespan` handler and reads its model, feature list and threshold from `models_artifacts/model_card.json` (no hard-coded algorithm).
- **Monitoring (plan D7):** `/metrics` (Prometheus) for service health; a per-feature **PSI drift report** vs the training reference; and delayed-label performance from the backtest. The December batch flags `frequency` (PSI 0.47), `interpurchase_interval_std` (0.45) and `purchase_frequency_trend` (0.38) — expected, since the Christmas trading window shifts cadence.
- **Quality gates:** ruff clean, mypy configured, **pytest with 75% coverage** on `src/`+`api/` (PRD requires ≥ 70%), GitHub Actions CI on every push.

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## 9. Mathematical Foundations Appendix

### 9.1 Logistic Regression & Binary Cross-Entropy

$P(y=1|x) = \sigma(w \cdot x + b)$, minimising $L = -(1/N) \sum [y \log \hat{y} + (1-y) \log(1-\hat{y})] + (\lambda/2)\|w\|^2$.

### 9.2 Isotonic Probability Calibration

Calibration finds a monotone $g$ minimising $\sum (y - g(\hat{y}))^2$ subject to $g$ non-decreasing, so that $P(\text{churn})$ can be multiplied by value to give an interpretable expected loss (Brier score).

### 9.3 Value-at-Risk & Budget-Constrained Targeting

$\text{VaR} = p \cdot v$. The campaign objective is the 0/1 knapsack $\max \sum x \, (s \cdot \text{VaR} - c) \text{ s.t. } \sum x \, c \le B$, solved by the greedy ratio $(s \cdot \text{VaR} - c)/c$; exact for equal costs and near-optimal otherwise, with a brute-force check.

### 9.4 BG/NBD + Gamma-Gamma (probabilistic CLV)

BG/NBD models the drop-out and purchase process; the conditional expectation of purchases in $(0, t]$ is $E[X(t) \mid x, t, T]$. Gamma-Gamma models the average order value. Their product is the expected value; $P(\text{alive})$ is the posterior drop-out survival probability.

### 9.5 K-Means & Silhouette

$\text{argmin}_S \sum \sum_{x \in S} \|x-\mu\|^2$; $s(i) = (b(i)-a(i)) / \max(a(i), b(i))$.

### 9.6 Evaluation Metrics

Precision $TP/(TP+FP)$, Recall $TP/(TP+FN)$, $F1 = 2PR/(P+R)$, $\text{ROC-AUC} = \int \text{TPR} \, d(\text{FPR})$, and top-decile value capture $= (\sum \text{value in top-ranked 10\%}) / (\sum \text{value})$.

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## 10. Why the numbers changed (random split → out-of-time)

The original draft used a single snapshot, a **random** 70/15/15 split, a 99-day window and an all-history cohort. That let train and test customers come from the same period and made AUC and $R^2$ look better than the model really is. The corrected, honest comparison:

| Dimension | Original draft | Corrected (this report) |
|---|---|---|
| Evaluation | Random stratified split (same period) | Out-of-time by cutoff; test used once |
| Churn test ROC-AUC | 0.7846 (XGBoost) | **0.7559** (Logistic Regression, selected on validation) |
| Churn recall | 81.3% | **75.8%** at the recall-constrained threshold |
| CLV test $R^2$ | 0.8447 (Ridge) | **0.4887** (LightGBM Tweedie); validation 0.7543 |
| Window | ~100 days | **90 days** (config-driven) |
| Cohort | 5,225 (all ever seen) | **4,323** active within 365 days |
| Churn rate | 55.25% | **50.45%** |

**Documented deviations and negatives (PRD §20.7):** 1. Out-of-time AUC is below the 0.80 target. This is expected: the task is harder when the model is scored on a later period, and the cohort excludes long-departed customers who made the old problem easy. The lower number is the honest one. 2. The original Ridge "$R^2$ 0.84" was a **leakage artifact** — `historical_clv` duplicated `monetary_total` and random splitting let a linear model exploit it. `historical_clv` was removed; the value model now beats a real baseline. 3. BG/NBD + Gamma-Gamma does not beat the supervised value model here (reported above). 4. Campaign save rates are assumptions; an A/B test design is the next step. 5. No real intervention data → no uplift modelling.

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## 11. Future Work

- Transformer sequence model (deferred by PRD §4.2).
- Real-time ingestion (Kafka/event pipeline) instead of scheduled batch scoring.
- Role-based access control and multi-tenant auth.
- Uplift modelling once A/B treatment data exists; measure the true save rate.
- Extend the shared engineering `template/` (P5) into FlowCast and Northgate.

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## 12. Verification Summary

| Check | Result |
|---|---|
| `docker compose up --build` on a clean clone | API :8000 + dashboard :8501 + populated DB |
| Out-of-time evaluation, test used once | Yes (asserted in `tests/test_data_split.py`) |
| 90-day windows enforced & tested | Yes (`tests/test_features.py`) |
| At-risk ranked by calibrated P(churn) × value | Yes API + dashboard |
| SHAP **and** LIME per customer in the dashboard | Yes `/customers/{id}/explain` |
| `/metrics`, drift report, Model Health page | Yes |
| Test coverage on `src/` (+`api/`) ≥ 70% | Yes 75% |
| Final report PDF with regenerated tables | Yes `docs/final_report.pdf` |

*— Vantara Retail Solutions Data Science Division —*